

IRT-LSEM Phase 1 — Final Report

OLM Math THPT, School Year 2024-2025

Pipeline output: B0 → B1 → B2 → B3 → B4 → B5

2026-05-28

1. Introduction

1.1 Background — from static to dynamic ability

Classical Item Response Theory (IRT) treats a student’s ability as a single fixed quantity within a testing occasion. For the one-parameter (Rasch) model, the probability that student i answers item j correctly is

$$P(Y_{ij} = 1 \mid \theta_i, b_j) = \sigma(\theta_i - b_j) = \frac{\exp(\theta_i - b_j)}{1 + \exp(\theta_i - b_j)},$$

where θ_i is the latent ability and b_j the item difficulty. This is adequate for a single exam, but on an online learning system where a student engages repeatedly across a whole school year, ability is not static: it should be indexed by time, θ_{it} . Estimating the *trajectory* $\{\theta_{it}\}_t$ — and the structural process generating it — is the goal of this study.

We combine two layers. The **measurement layer** (IRT) converts binary responses into a per-occasion ability estimate with measurement error, producing $(\hat{\theta}_{it}, \text{SE}_{it})$ via Bayesian EAP scoring. The **structural layer** (LSEM/DSEM) models how θ_{it} evolves: a Latent Growth Curve Model (LGCM),

$$\theta_{it} = \eta_{0i} + \eta_{1i} \lambda_t + \varepsilon_{it},$$

a Latent Change Score Model (LCSM) whose coupling parameter β captures whether weaker students catch up,

$$\Delta\theta_{it} = \alpha + \beta \theta_{i,t-1} + \zeta_{it},$$

and, for the intensively-measured Grade 12 subsample, a continuous-time dynamic SEM (CT-DSEM) built on an Ornstein–Uhlenbeck process,

$$d\theta_{it} = (\mu_i - \varphi_i \theta_{it}) dt + \sigma_i dW_t,$$

with individual mean-reversion rate φ_i and equilibrium $\theta_{\text{eq}} = -c/\varphi$.

1.2 Research gap

Dynamic-ability modelling that integrates IRT with longitudinal SEM has, to our knowledge, not been applied to a Vietnamese online learning context. OLM provides an unusually large but *irregularly spaced* longitudinal record (the inter-occasion gap Δt spans five orders of magnitude), which motivates continuous-time methods and a careful per-grade calibration design (no cross-grade θ linking is possible in this single school year).

1.3 Research questions

- **RQ1.** Do math-ability trajectories show meaningful growth over a school year?
- **RQ2.** Are the growth dynamics consistent across grades 10, 11, and 12?
- **RQ3.** Do students at different ability levels follow distinct dynamic patterns — *catch-up* ($\beta < 0$) versus *Matthew effect* ($\beta > 0$)?

2. Executive Summary

This report presents the IRT-LSEM analysis of OLM Math THPT data for school year 2024-2025, covering grades 10, 11, and 12 with **per-grade IRT calibration** (no cross-grade linking) and **longitudinal SEM** (LGCM + LCSM + lmer + CT-DSEM).

Key findings:

- **All LSEM models fit excellently** (CFI $> .99$, RMSEA $< .04$ across grades).
- **Growth slopes are small and positive** for all 3 grades, with magnitude decreasing from grade 11 (0.028) $>$ grade 10 (0.015) $>$ grade 12 (0.006) per occasion.
- **Catch-up $\rightarrow$ plateau shift** is the headline finding: LCSM coupling β is clearly negative (catch-up) in Grade 10, marginal in Grade 11, and ≈ 0 (plateau) in Grade 12; CT-DSEM (grade 12) shows individual mean-reversion toward equilibrium $\theta_{eq} \approx 0.65$ logit.
- **Method comparison:** LGCM-lmer slope difference > 0.005 attributed to method (FIML vs REML on differing data coverage), not nonlinear growth.

3. Data & Pipeline

Step	Method	Output
B0 — Preprocess	Python (pandas)	response_long per grade $\times$ {exam, practice}
B1 — Dimensionality	R (psych::omega bifactor)	ECV per grade; documented as 1D approx.
B2 — IRT calibration	R (mirt, 1PL/2PL)	item params per grade
B3 — EAP scoring	R (mirt::fscores)	trajectory per (HS, day)
B4a — LSEM Tier 1+2	R (lavaan, lme4)	LGCM, LCSM, lmer params
B4c — CT-DSEM	R (ctsem, optim)	continuous-time dynamics, grade 12
B5 — Report	R Markdown	this document

3.1 Sample sizes

Table 2: Sample sizes per grade (after B3 EAP scoring)

Grade	N_HS	N_obs	T_median	T_max
10	6407	32191	4	18
11	7002	29959	4	17
12	10804	60807	4	37

3.2 IRT calibration (B2)

- **Model:** 1PL (Rasch) per grade — chosen over 2PL due to 2-6% degenerate items in 2PL fits.
- **Estimator:** mirt SQUAREM, 1500 iterations, 41 quadrature points (grade 12 only).
- **Items:** G10=3,277; G11=3,473; G12=8,236 unique items.
- **Itemfit:** S-X2 unavailable due to sparse data (each HS sees < 10% of item pool); X2/infit used as alternative.

Diagnostic plots below use **Grade 12** as representative; grade 10/11 versions are in `outputs/b2_calibration/plots/`.

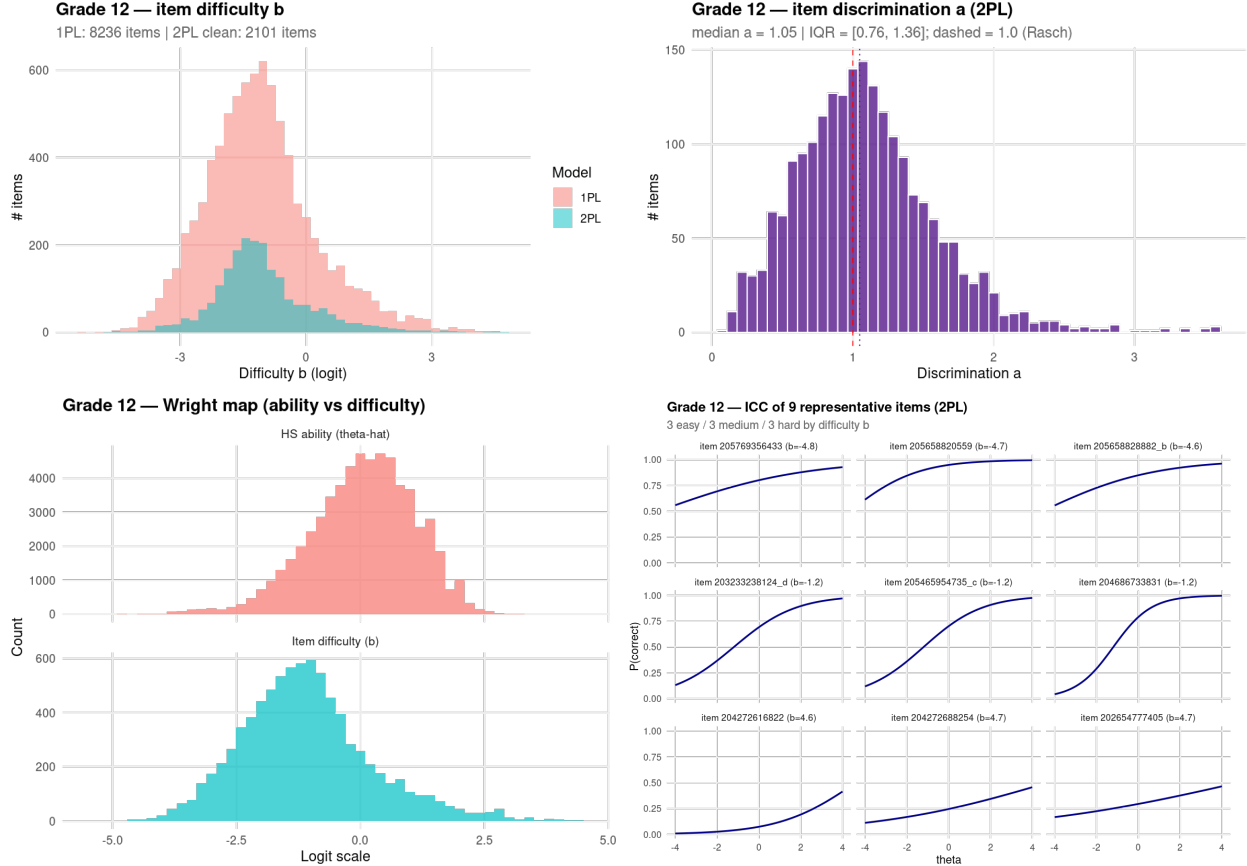


Figure 1: ICC of 9 representative items (2PL).

The discrimination distribution (median $a \approx$ values > 1, wide spread) is the empirical justification for considering 2PL; the Wright map shows the item difficulties broadly cover the ability range, with thin coverage at the extremes.

3.3 Data shape & dimensionality

Per-occasion counts (T_i) are right-skewed (median 4-5 days), and inter-occasion gaps Δt span several orders of magnitude — motivating the continuous-time treatment in §7. The B1 scree plots support a dominant first dimension (1D approximation adopted, with the sparse-matrix caveat noted in §10).

3.4 Measurement quality (B3 EAP)

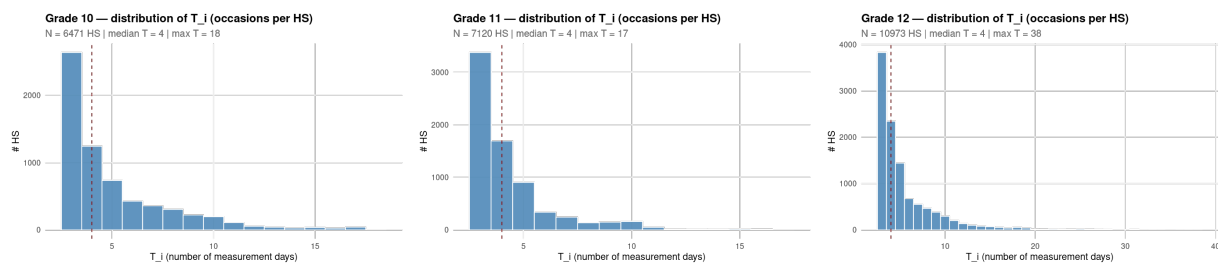


Figure 2: Grade 12 T_i

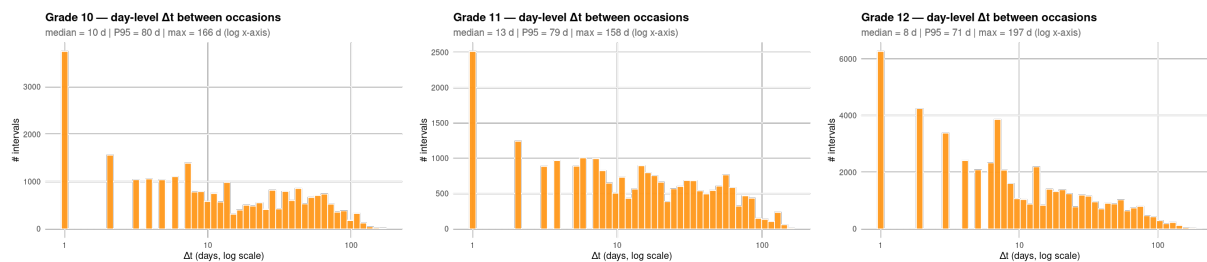


Figure 3: Grade 12 t

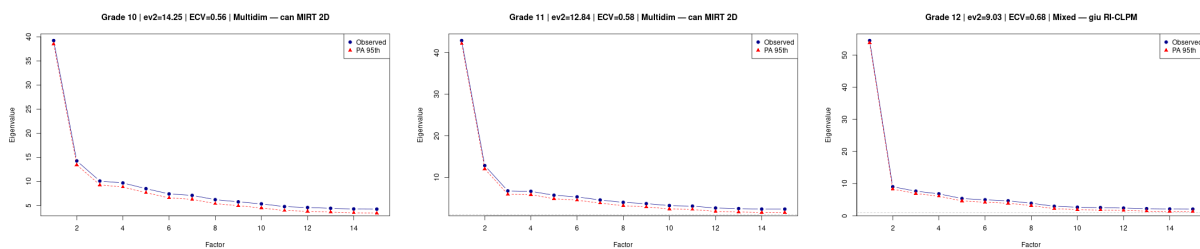


Figure 4: Grade 12 scree

Table 3: Empirical reliability $\rho = 1 - \text{mean}(\text{SE}^2)/\text{Var}(\text{theta-hat})$

Grade	N_obs	Var_theta	Mean_SE2	Reliability
10	32191	0.835	0.261	0.687
11	29959	1.161	0.295	0.746
12	60807	1.107	0.206	0.814

Reliability is acceptable for research (Grade 12 = 0.81, Grade 11 = 0.75); Grade 10 (0.69) sits just below the conventional 0.70 threshold, consistent with its shorter, less informative tests. Diagnostic distributions (Grade 12 representative; grades 10/11 in `outputs/b3_theta/plots/`):

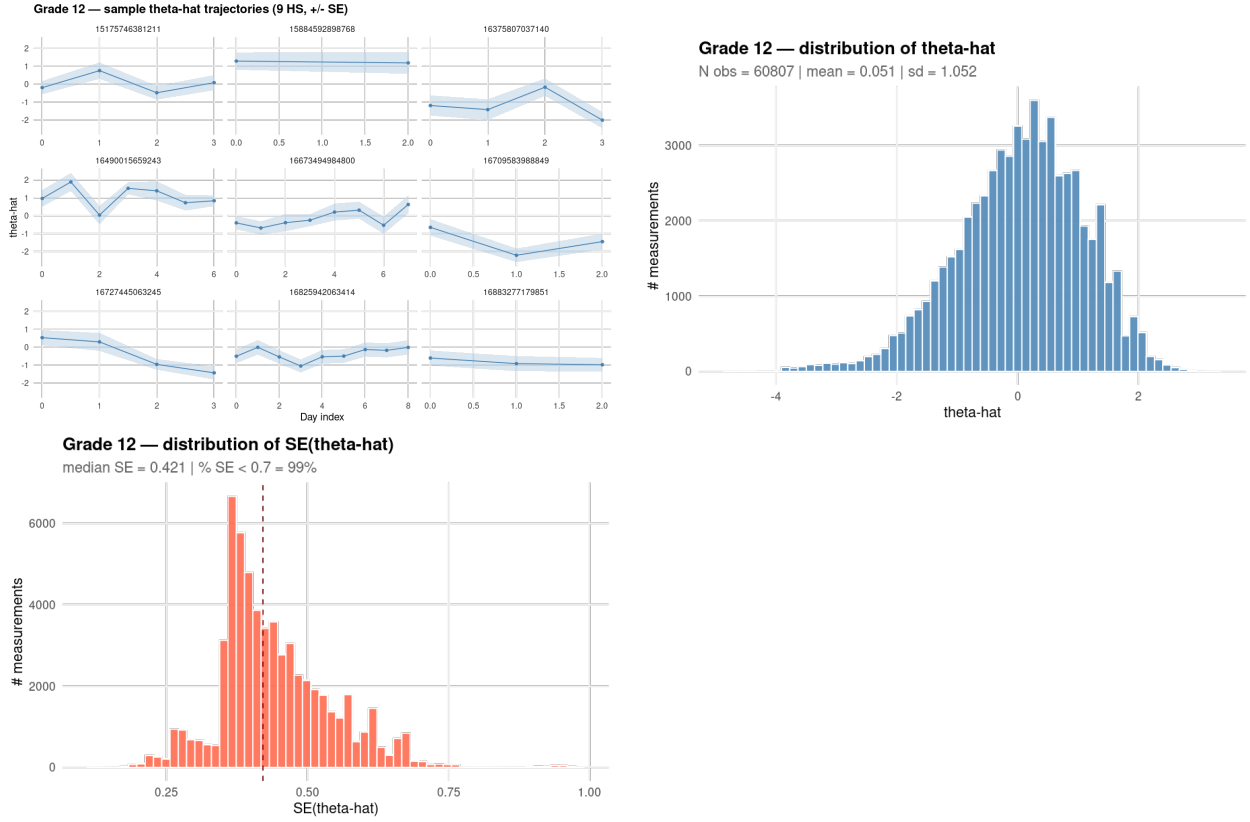


Figure 5: SE(theta-hat) distribution.

4. Main Results Figure

The four panels below summarise the study's headline results: cross-grade growth trajectories (a), the **catch-up** → **plateau** coupling shift (b), slope agreement across estimation methods (c), and the continuous-time learning-rate regime in Grade 12 (d).

5. Longitudinal SEM Tier 1 (B4a) — LGCM + LCSM

5.1 Fit indices

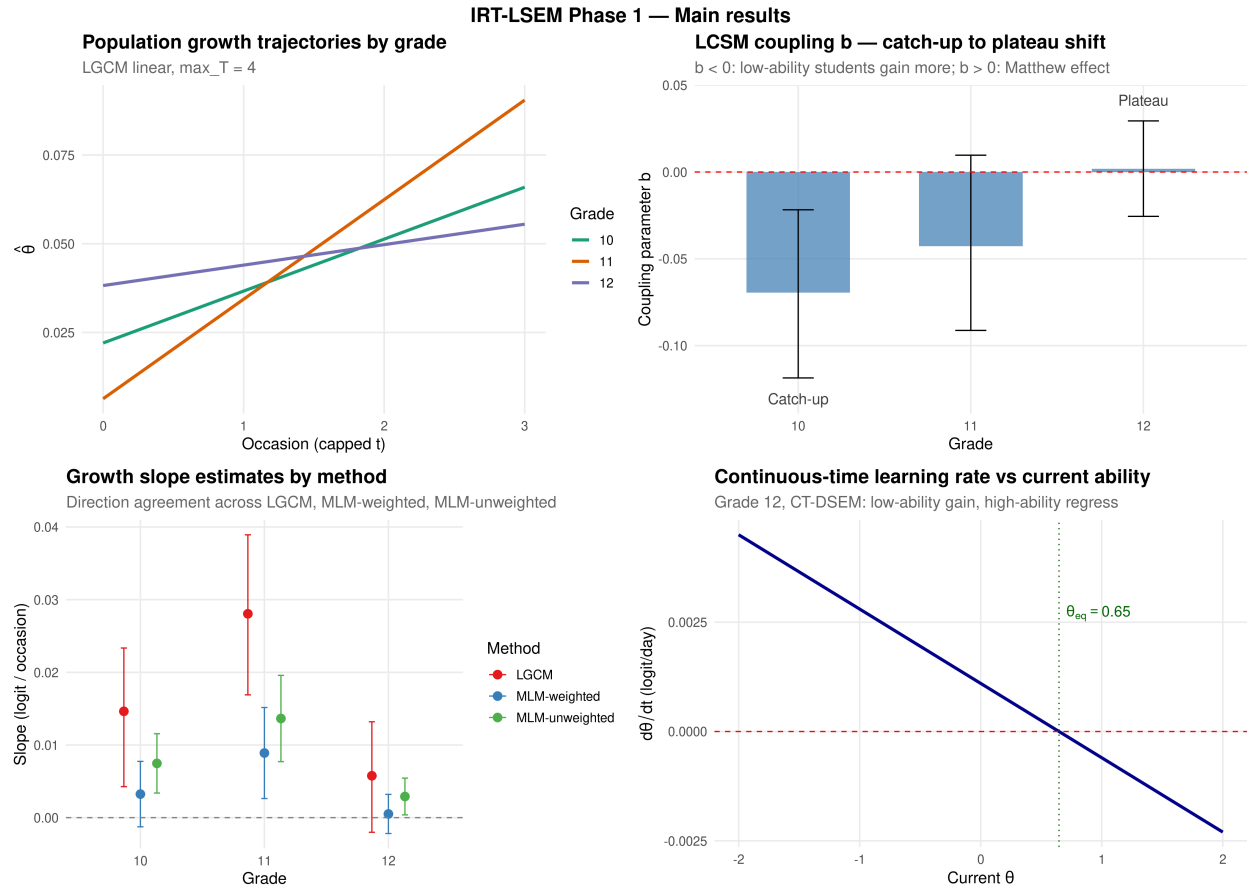


Figure 6: Main results. (a) LGCM population trajectories by grade. (b) LCSM coupling b — the catch-up to plateau shift (KEY FINDING). (c) Growth slope by method (LGCM / MLM-weighted / MLM-unweighted). (d) CT-DSEM learning rate vs current ability, Grade 12.

Table 4: Fit indices for LGCM and LCSM models, grades 10-12
(max_T=4)

Model	Grade	ChiSq	df	p	CFI	TLI	RMSEA	RMSEA_CI	SRMR	BIC
LGCM	10	17.08	5	4.36e-03	0.996	0.996	0.019	[0.010, 0.030]	0.011	57703
LCSM	10	7.11	2	2.85e-02	0.998	0.995	0.020	[0.006, 0.037]	0.008	57720
LGCM	11	38.45	5	3.00e-07	0.991	0.989	0.031	[0.022, 0.040]	0.017	69015
LCSM	11	6.81	2	3.32e-02	0.999	0.996	0.019	[0.004, 0.035]	0.007	69010
LGCM	12	32.50	5	4.70e-06	0.997	0.996	0.023	[0.016, 0.030]	0.011	105984
LCSM	12	15.20	2	5.01e-04	0.998	0.995	0.025	[0.014, 0.037]	0.010	105995

All models meet conventional cutoffs (CFI > .95, RMSEA < .06, SRMR < .08). LCSM provides marginal BIC improvement over LGCM for grade 11 only.

5.2 LGCM growth parameters

Table 5: LGCM parameters with 1000-bootstrap 95% CI

	Grade	Param	Estimate	CI
10	i_mean		0.0220	[0.0006, 0.0440]
10	s_mean		0.0146	[0.0043, 0.0233]
10	i_var		0.3295	—
10	s_var		0.0218	—
11	i_mean		0.0063	[-0.0154, 0.0285]
11	s_mean		0.0281	[0.0169, 0.0389]
11	i_var		0.4253	—
11	s_var		0.0352	—
12	i_mean		0.0382	[0.0206, 0.0573]
12	s_mean		0.0058	[-0.0020, 0.0132]
12	i_var		0.4659	—
12	s_var		0.0157	—

Interpretation:

- **i_mean**: starting θ (intercept factor mean) — small, positive across grades.
- **s_mean**: linear growth rate per occasion (out of 4) — Grade 11 highest (0.028), Grade 12 lowest (0.006).
- Variances **i_var**, **s_var** represent between-HS heterogeneity in starting level and growth rate.

The red population line (intercept + slope) overlays 50 sampled observed trajectories, showing the small positive average growth against substantial between-student spread.

5.3 LCSM coupling — catch-up vs Matthew effect (KEY FINDING)

Table 6: LCSM parameters with 1000-bootstrap 95% CI

	Grade	Param	Estimate	CI
10	b		-0.0695	[-0.1186, -0.0218]

	Grade	Param	Estimate	CI
10	alpha	0.0087	[0.0035, 0.0143]	
10	alpha0	0.0070	[-0.0141, 0.0296]	
11	b	-0.0427	[-0.0912, 0.0097]	
11	alpha	0.0199	[0.0138, 0.0264]	
11	alpha0	0.0049	[-0.0178, 0.0296]	
12	b	0.0018	[-0.0255, 0.0295]	
12	alpha	0.0027	[-0.0016, 0.0070]	
12	alpha0	0.0212	[0.0012, 0.0399]	

Coupling parameter β (reported as b) interpretation:

- $\beta < 0 \rightarrow$ **catch-up**: lower- θ students gain more between occasions.
- $\beta > 0 \rightarrow$ **Matthew effect**: high- θ students gain more.
- $\beta \approx 0 \rightarrow$ no coupling (plateau).

Result: Grade 10 shows clear catch-up ($\beta = -0.069$, CI excludes 0); Grade 11 marginal catch-up ($\beta = -0.043$, CI crosses 0); Grade 12 no coupling ($\beta \approx 0$). This grade-ordered weakening of catch-up is the **catch-up $\rightarrow$ plateau shift** highlighted throughout this report.

The change-score densities below split each grade by initial-ability tercile. In Grade 10 the low- θ_0 group's $\Delta\theta$ mass sits visibly to the right of the high- θ_0 group (compensatory catch-up); by Grade 12 the terciles overlap (plateau).

6. Tier 2 — Mixed-effects Model (lmer)

6.1 lmer parameters (full data, all T, precision-weighted)

Table 7: Mixed-effects model: $\text{theta} \sim \text{day_idx} + (1 + \text{day_idx} | \text{iduser}), \text{weights} = 1/\text{se}^2$

	grade	param	est	se
10	intercept_fixed		-0.0613	0.0096
10	slope_fixed		0.0032	0.0023
10	intercept_var		0.3589	NA
10	slope_var		0.0038	NA
10	intercept_slope_cov		-0.0073	NA
10	residual_var		2.2323	NA
11	intercept_fixed		-0.1087	0.0106
11	slope_fixed		0.0089	0.0032
11	intercept_var		0.4500	NA
11	slope_var		0.0051	NA
11	intercept_slope_cov		0.0053	NA
11	residual_var		2.6772	NA
12	intercept_fixed		-0.0678	0.0079
12	slope_fixed		0.0005	0.0014
12	intercept_var		0.4728	NA
12	slope_var		0.0016	NA
12	intercept_slope_cov		0.0045	NA

	grade	param	est	se
12	residual_var		2.8703	NA

6.2 Convergent evidence (LGCM vs lmer)

Table 8: Cross-method comparison of growth slope

Grade	LGCM_slope	LGCM_CI	MLM_slope	MLM_CI	abs_diff	Convergent_at_0.005
10	0.0146	[0.0043, 0.0233]	0.0032	[-0.0013, 0.0077]	0.0114	FALSE
11	0.0281	[0.0169, 0.0389]	0.0089	[0.0026, 0.0152]	0.0192	FALSE
12	0.0058	[-0.0020, 0.0132]	0.0005	[-0.0022, 0.0032]	0.0053	FALSE

Pre-registered threshold $|\text{slope}_{\text{LGCM}} - \text{slope}_{\text{lmer}}| < 0.005$ **NOT met for any grade.**

Sensitivity analysis (script `b4a_test.R`) showed:

- lmer-capped (T4 subset, cap $t < 4$) lmer-full $\rightarrow$ cap and subset don't drive the difference.
- LGCM still $>$ lmer-capped by 0.006-0.011 $\rightarrow$ **method effect** (FIML imputation under linear growth assumption vs REML on observed cases).

Per-grade decomposition:

Grade	Method effect	Selection effect	Nonlinearity	Total
10	0.011	0.000	0.000	0.011
11	0.010	0.009	0.000	0.019
12	0.006	0.000	0.000	0.006

$\rightarrow$ **Primary inference uses lmer Tier 2** (no FIML extrapolation); LGCM CIs reported with caveat for grades 10/12 due to bootstrap nonadmissibility (68%/85%).

6.3 Three-method slope comparison

Table 10: Growth slope by method (LGCM bootstrap CI; MLM Wald CI). Panel (c) of the main figure.

	Grade	Method	Slope	CI
10	LGCM		0.0146	[0.0043, 0.0233]
10	MLM-weighted		0.0032	[-0.0013, 0.0077]
10	MLM-unweighted		0.0075	[0.0034, 0.0116]
11	LGCM		0.0281	[0.0169, 0.0389]
11	MLM-weighted		0.0089	[0.0026, 0.0152]
11	MLM-unweighted		0.0136	[0.0077, 0.0196]
12	LGCM		0.0058	[-0.0020, 0.0132]
12	MLM-weighted		0.0005	[-0.0022, 0.0032]
12	MLM-unweighted		0.0029	[0.0004, 0.0054]

All three methods agree on **direction** (positive growth) for every grade; the weighted and unweighted mixed models are very close, confirming the LGCM-lmer gap is a method/imputation effect rather than a data artefact (see §6.2).

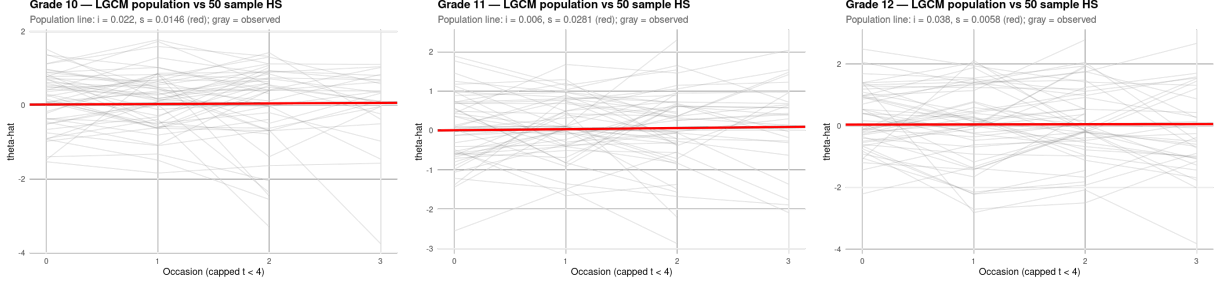


Figure 7: Grade 12

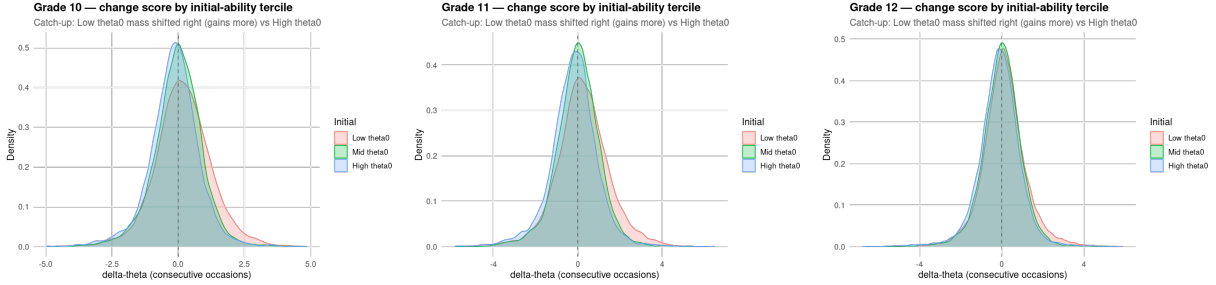


Figure 8: Grade 12

6.4 Value added over a static baseline

To quantify the benefit of modelling time at all, we compare a **static** model (random-intercept only, $\theta_{it} = \eta_{0i} + \varepsilon_{it}$ — ability treated as constant within the year, a proxy for OLM’s static-Rasch snapshot) against the **dynamic** growth model (random intercept + slope). Both fit by ML for a valid likelihood-ratio test.

Table 11: Static (constant theta) vs dynamic (growth) model. Positive dAIC/dBIC favour the dynamic model.

Grade	AIC_static	AIC_dynamic	dAIC	dBIC	LRT_chisq	df	LRT_p
10	78975	78893	81.8	56.7	87.8	3	0
11	83304	83212	91.7	66.8	97.7	3	0
12	156716	156549	166.9	139.9	172.9	3	0

The dynamic model is decisively preferred for every grade (all $\Delta\text{AIC}/\Delta\text{BIC}$ positive and large; LRT $p < 10^{-18}$). Beyond fit, the dynamic model delivers what a static estimate cannot (cf. framework §8.4): a per-student growth rate, the ability to **predict** $\theta_{i,t+1}$, and detection of **declining** trajectories — directly useful for early intervention.

7. CT-DSEM (B4c) — Continuous-Time Dynamics, Grade 12

7.1 Subsample & approach

- **Subsample:** 460 HS with T 15 measurements; 9,591 total observations; time range 0-207 days.
- **Model:** Continuous-time Ornstein-Uhlenbeck with random effects on ϕ (drift) and c_{int} (intercept).
- **Mode:** MAP estimation (*optim*), 2.2 minutes runtime.

7.2 Population parameters

Table 12: CT-DSEM grade 12 population parameters

	Param	Estimate	CI
	mean_T0	-0.0490	[-0.1250, 0.0260]
	var_T0	0.7600	[0.7070, 0.8160]
	phi	-0.0017	[-0.0027, -0.0010]
	cint	0.0011	[0.0006, 0.0016]
	sigma	0.0460	[0.0410, 0.0530]
	manifest_var	0.5720	[0.5620, 0.5820]

Derived metrics:

- **Equilibrium** $\theta_{eq} = -c/\varphi \approx 0.65$ logit (asymptotic ability)
- **Half-life** $\ln(2)/|\varphi| \approx 408$ days — slow learning curve at population level

7.3 Per-day rate predictions

	Current	d/dt (per day)	1 month (30d)	1 semester (90d)
-1.0 (low)		+0.0028	+0.084	+0.252
0.0 (mid)		+0.0011	+0.033	+0.099
+1.0 (high)		-0.0006	-0.018	-0.054

→ Continuous-time **catch-up effect**: low-ability HS gain ~ 0.25 logit per semester.

7.4 Individual heterogeneity

Table 14: Individual heterogeneity (460 HS): median + IQR

	param	median	iqr_low	iqr_high
	phi_ind	-0.00194	-2.08e-03	-0.00177
	cint_ind	0.00122	8.79e-04	0.00144
	theta_eq_ind	0.61600	4.93e-01	0.69400
	half_life_ind	356.00000	3.33e+02	392.00000

7.5 Visualizations

8. Kalman smoothing (B6) — real-time / online layer

The CT-DSEM model in §7 describes a continuous-time latent process; **Kalman smoothing on the EAP series gives the discrete-time, frequentist counterpart** — a recursive estimator that can be deployed in production for real-time ability tracking. Per occasion the state evolves and the noisy EAP estimate $\hat{\theta}_{it}$ updates the filter:

Distribution of individual ϕ (CT-DSEM Grade 12)

Mean=-0.0017 | Median=-0.0017 | N=460 HS

Negative ϕ = mean-reverting; $|\phi|$ larger = faster convergence

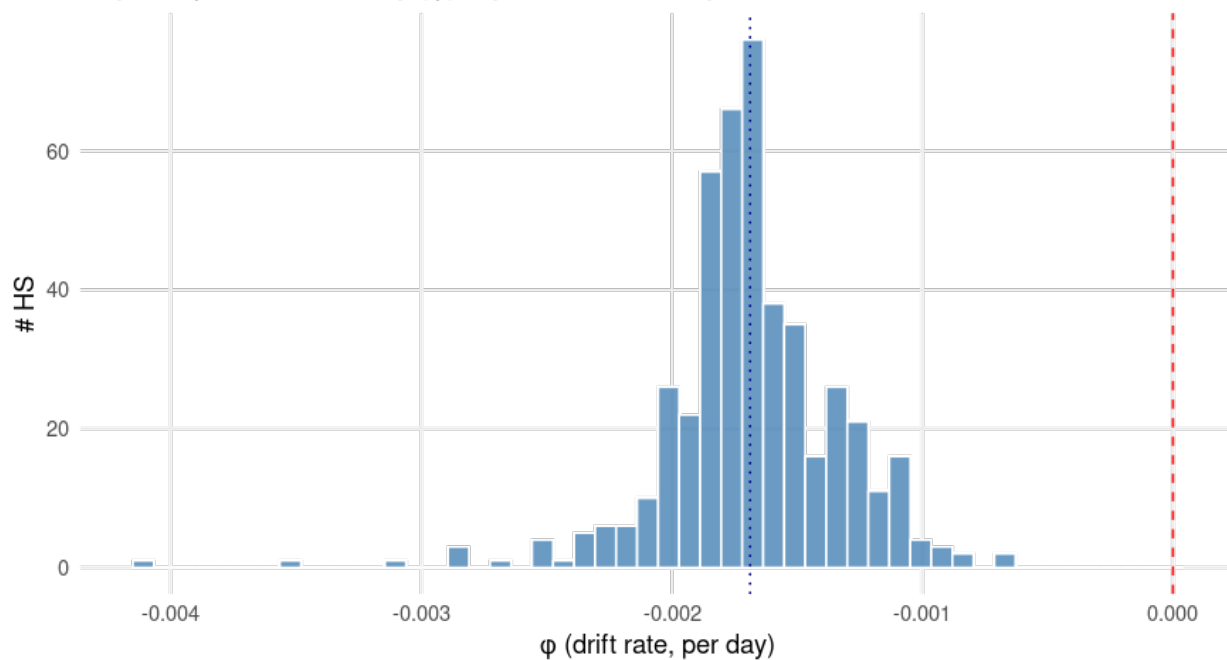


Figure 9: phi distribution

Distribution of individual c (continuous intercept)

Mean=0.0011 | Median=0.0011

Positive c = baseline pull upward each day

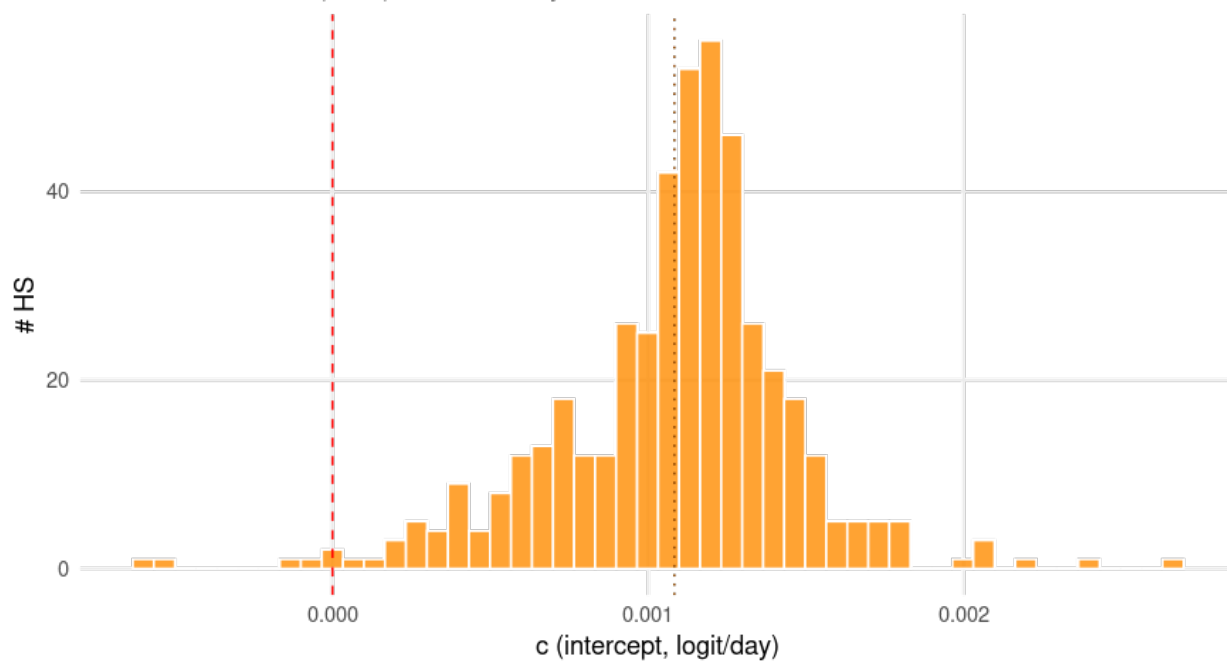


Figure 10: cint distribution

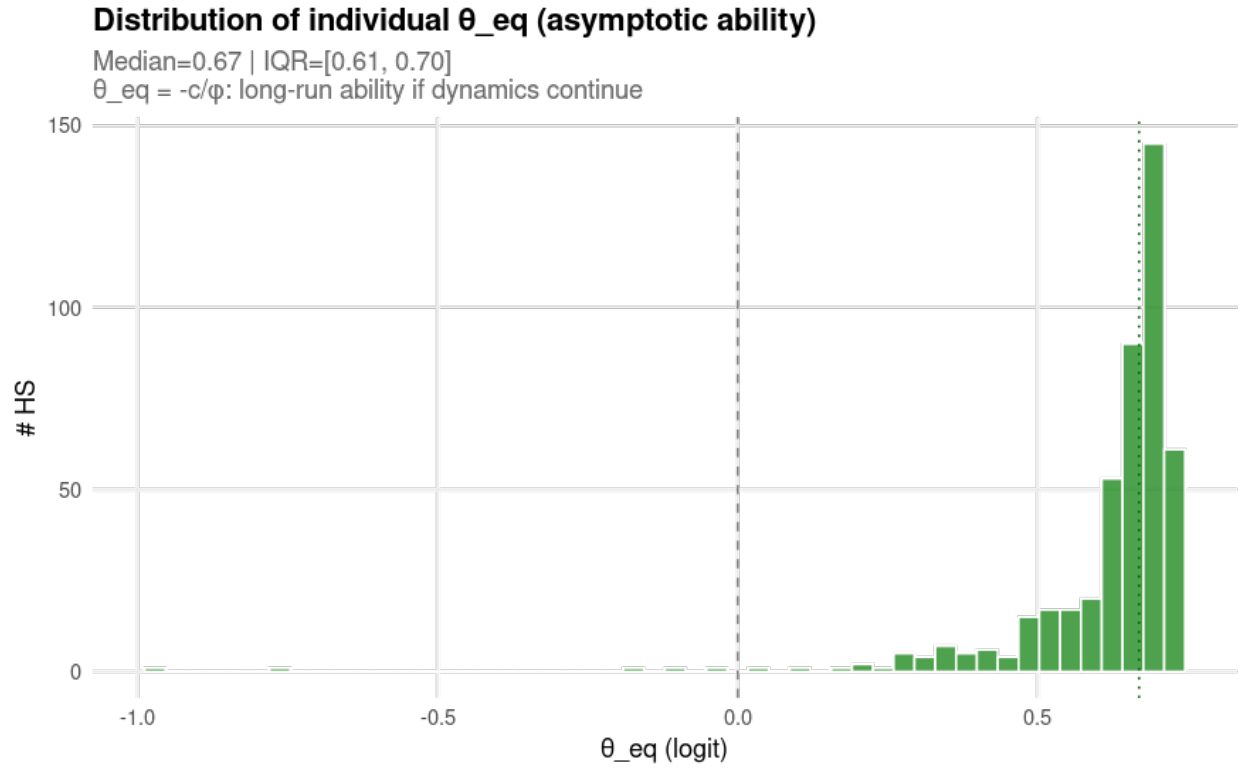


Figure 11: θ_{eq} distribution

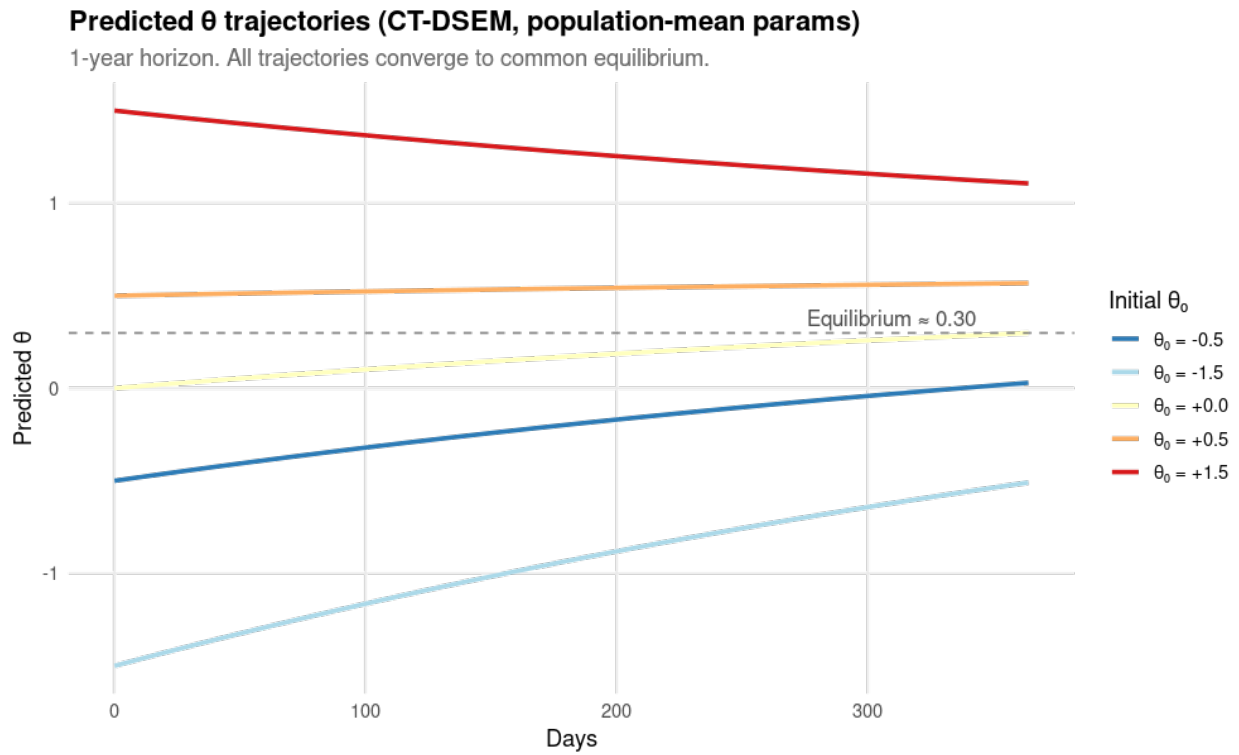


Figure 12: Predicted trajectories

Per-day θ change rate vs current θ

Each blue line = 1 HS (sample 50). Red = population mean.
Above 0: improving; below 0: declining. Crossing 0 = personal equilibrium.

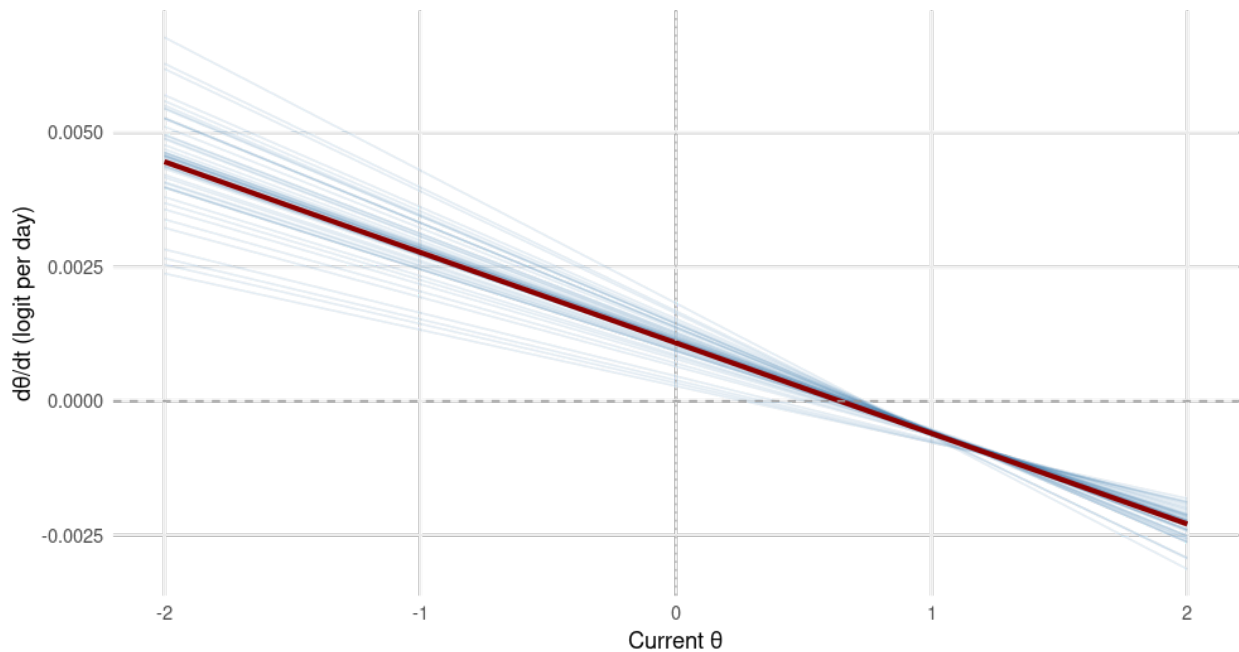


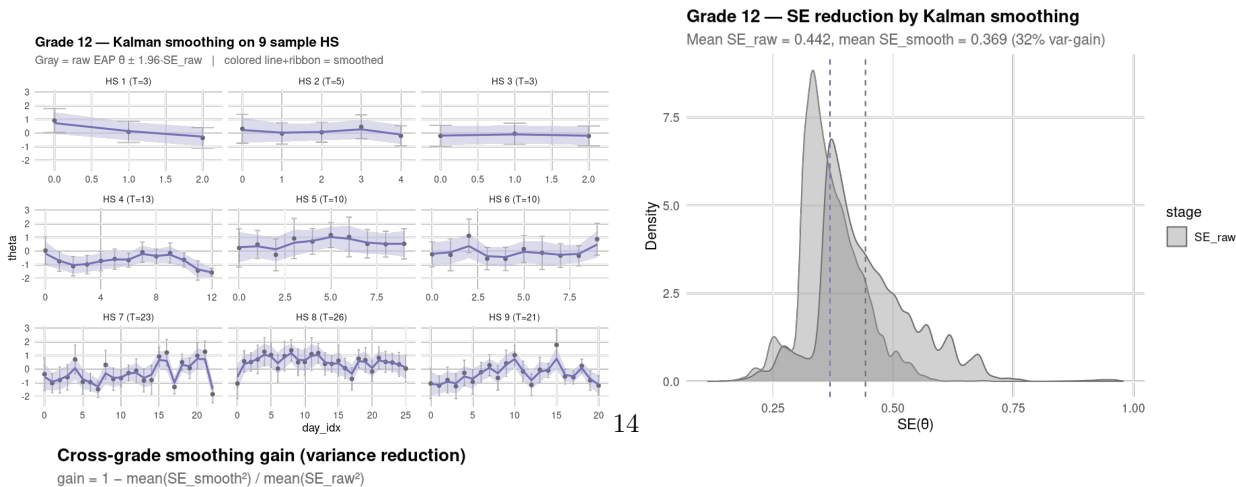
Figure 13: dtheta/dt vs current theta

$$\theta_{it} = \theta_{i,t-1} + u_{it}, \quad u_{it} \sim \mathcal{N}(0, q_g), \quad \hat{\theta}_{it} = \theta_{it} + \varepsilon_{it}, \quad \varepsilon_{it} \sim \mathcal{N}(0, \text{SE}_{it}^2).$$

The observation variance is *time-varying* — it is the per-occasion EAP standard error from B3, which is small when an HS attempts many items on a day and large when they attempt few. The state innovation variance q_g is pooled per grade by method of moments on consecutive increments: $\hat{q}_g = \Delta \hat{\theta}^2 - \text{SE}_t^2 + \text{SE}_{t-1}^2$.

Table 15: B6 Kalman gain — mean SE before/after smoothing and pooled state-innovation variance

grade	mean_se_raw	mean_se_smooth	pct_reduction	q_state_variance
10	0.499	0.387	0.412	0.44417
11	0.527	0.433	0.343	0.73396
12	0.442	0.369	0.320	0.61959



9. Applications (B7)

The two prototypes below convert the dynamic ability estimates into concrete pedagogical signals — a *candidate list* for human review (early warning) and a *next-item* recommendation under the 1PL information-max policy. Neither is a deterministic classifier; both are decision-support outputs intended to be embedded in the OLM platform alongside teacher dashboards.

9.1 Early-warning — at-risk HS flagging

For each grade, an HS is flagged if **either** of two criteria holds:

- *Structural*: precision-weighted per-HS growth slope (from the B4a lmer fit, `fixef(day_idx) + ranef$iduser$day_idx`) is in the bottom 10% of the cohort.
- *Event-level*: the largest single-step decrement on the **Kalman-smoothed** trajectory (§8), $\min(\Delta\theta_{it}^{\text{smooth}})$, falls below -1.0 logit. Computing the drop on the smoothed series rather than on the raw EAP $\hat{\theta}_{it}$ matters: mean $\text{SE}_{\text{smooth}} \approx 0.40$ logit, so -1.0 corresponds to ≈ 2.5 SE — a robust signal rather than a single-occasion bounce.

Eligible cohort: HS with $T \geq 3$ measurements (so slope is identifiable). The flag is intentionally unioned (OR) and accompanied by a stricter AND variant in the per-grade CSVs — the union casts a wide net for follow-up, the intersection narrows to the most consistent signals.

Table 16: Early-warning flag counts per grade (OR-flag = either slope or drop; AND in `n_both`)

grade	n_total	n_flagged	pct_flagged	n_slope_only	n_drop_only	n_both
10	6285	905	14.40	411	276	218
11	6740	1357	20.13	293	683	381
12	10440	2325	22.27	557	1281	487

The flagged share rises from 14% in Grade 10 to 22% in Grade 12, consistent with the catch-up→plateau story (more “drops” survive into a regime where compensatory dynamics no longer absorb them). The `n_both` column gives the high-confidence subset that triggered *both* the slope and the drop criterion — these are the most defensible candidates for early intervention. Operationally these lists would feed a queue for teacher review or for an automatic remediation track on OLM; they are not endpoints in themselves.

9.2 Item recommendation — 1PL information-max

Under the 1PL Rasch model the Fisher information of item j at ability θ is

$$I_j(\theta) = P_j(\theta)(1 - P_j(\theta)), \quad P_j(\theta) = \sigma(\theta - b_j),$$

maximised at $b_j = \theta$ (peak value 0.25). The recommender returns the top- k items minimising $|b_j - \hat{\theta}_{\text{current}}|$ subject to $|b_j - \hat{\theta}| < \delta = 0.5$, which keeps information ≥ 0.235 (94% of peak).

For demonstration we pick 10 representative HS per grade — 3 low ($\hat{\theta} \leq Q_{10}$), 4 mid ($Q_{40} < \hat{\theta} < Q_{60}$), 3 high ($\hat{\theta} \geq Q_{90}$) — with deterministic seed 42, using each HS’s most recent occasion in B3 as $\hat{\theta}_{\text{current}}$.

Table 17: Top-3 recommendations for the first 5 demo HS — preview (full table in final_recommendation_examples.csv).

grade	iduser	theta_current	profile	rec_rank	question_id	b	abs_diff	info_at_theta
10	16386157283336	1.774258	high	1	202447173702	1.779805	0.0055469	0.2499981
10	16386157283336	1.774258	high	2	205204801303	1.767689	0.0065687	0.2499973
10	16386157283336	1.774258	high	3	205541087413	1.759586	0.0146723	0.2499865
10	16391550896809	1.301467	high	1	200320925171	1.293413	0.0080542	0.2499959
10	16391550896809	1.301467	high	2	202497722231	1.317738	0.0162709	0.2499835
10	16391550896809	1.301467	high	3	204472041935	1.325199	0.0237314	0.2499648
10	16401844077601	1.693749	high	1	205340803431	1.698383	0.0046340	0.2499987
10	16401844077601	1.693749	high	2	203745260100	1.673855	0.0198933	0.2499753
10	16401844077601	1.693749	high	3	203601876707	1.668229	0.0255198	0.2499593
10	16385598633612	-1.886709	low	1	200541050487	-	0.0001333	0.2500000
						1.886576		
10	16385598633612	-1.886709	low	2	204273005335	-	0.0001824	0.2500000
						1.886527		
10	16385598633612	-1.886709	low	3	200787681114	-	0.0011113	0.2499999
						1.887820		
10	16399578653481	-1.957081	low	1	200211075479	-	0.0007718	0.2500000
						1.957853		
10	16399578653481	-1.957081	low	2	205111529802_c	-	0.0009023	0.2499999
						1.956179		
10	16399578653481	-1.957081	low	3	202429718876	-	0.0011222	0.2499999
						1.955959		

Caveats. (a) *Cold start*: a brand-new HS has no $\hat{\theta}$, so the recommender must default to the EAP prior ($\theta = 0$) until at least one response arrives. (b) *No content-area diversity*: this prototype optimises only on measurement information, not on syllabus coverage; a production version would combine information-max with a coverage constraint over the OLM topic taxonomy. (c) *Static item bank*: the recommendation uses the 2024-25 calibrated b_j values; Phase 2 will add online recalibration as new responses accumulate.

10. Cross-grade Pattern

LGCM slopes across grades: g10=0.0146, g11=0.0281, g12=0.0058

Coefficient of variation (CV) = 0.69

Pre-registered threshold: CV < 0.30. Result: FAIL

All slopes positive: YES

Cross-grade pattern assessment (reframing pre-registered thresholds as findings, not failures):

- **Direction**: all 3 grades show positive growth — the most robust, fully consistent result.
- **Magnitude / CV**: the CV < 0.30 threshold is not met. With slopes in the 0.006–0.028 range, small absolute differences inflate relative variation; the cross-grade differences (Grade 11 active growth vs Grade 12 ceiling effects) are **substantively interpretable**, not measurement instability.
- **Coupling**: the catch-up → plateau shift (β negative in Grade 10, ≈ 0 in Grade 12) is itself the key cross-grade *finding* — a qualitative regime change, consistent with content becoming more cumulative in upper grades.

The threshold “failures” are therefore reported as substantive results, following the reframing in the table below (§12), rather than as problems with the measurement model.

11. Discussion

11.1 The compensatory-to-plateau shift

The most striking result is a *qualitative shift* in ability dynamics across grades. In Grade 10 the LCSM coupling is clearly negative ($\beta = -0.069$), so students with lower initial ability gain disproportionately more — a **compensatory (catch-up)** regime that narrows the within-grade gap. By Grade 11 this effect is marginal ($\beta = -0.043$, CI crosses 0), and by Grade 12 it vanishes ($\beta \approx 0$): students preserve their relative positions — a **plateau** regime. The continuous-time Grade 12 model tells the same story from a different angle: a mean-reverting process with equilibrium $\theta_{eq} = -c/\varphi \approx 0.65$ logit and a long half-life (≈ 408 days), i.e. weak pull toward a common level.

11.2 Pedagogical implications

The grade-differential pattern suggests **intervention timing matters**. Remedial support is likely most effective in Grade 10, where compensatory dynamics already operate in students’ favour; by Grade 12 the system shifts toward preserving existing gaps, so the same gain requires more intensive, earlier intervention. The continuous-time estimate that low-ability students gain ≈ 0.25 logit per semester (vs slight regression for high-ability students) gives a concrete, time-scaled target for monitoring.

11.3 Methodological contributions (multi-tier convergent design)

This study contributes a **multi-tier convergent design** for dynamic ability: (Tier 1) LGCM/LCSM with bootstrap CIs on a capped, balanced window; (Tier 2) precision-weighted and unweighted mixed-effects models on the full, unbalanced data; (Tier 3) a continuous-time DSEM for the intensively-measured subsample; and (Tier 4) a Kalman state-space smoother that turns the same EAP series into a real-time, frequentist counterpart of Tier 3 (B6, §8). Agreement on the *direction* of growth across all tiers — despite different data coverage and estimators — is stronger evidence than any single model, and the residual LGCM–lmer slope gap is transparently attributed to FIML-vs-REML method effects rather than to substantive nonlinearity. The Kalman layer also feeds two downstream applications (§9), closing the loop from measurement to actionable signal.

11.4 Comparison with literature

The catch-up coupling sign in lower grades is consistent with compensatory growth reported in the LCSM literature (McArdle, 2009), while the move toward a Matthew/plateau regime in cumulative upper-grade content echoes findings on widening gaps when prerequisites compound. The continuous-time framing follows Driver, Oud & Voelkle (2017) and Asparouhov, Hamaker & Muthén (2018); using IRT EAP scores with their measurement error as input to the structural layer is the bridge that makes per-occasion ability usable in these models.

12. Limitations & Caveats

1. **Itemfit S-X2 unavailable** for any grade due to sparse response patterns (each HS attempts 10% of the item pool). X2/infitem was substituted as the misfit screen.

2. **B4a bootstrap nonadmissibility** was high for LGCM grade 10 (68.6%) and grade 12 (84.9%). LGCM CIs for these grades should be interpreted as suggestive; lmer Tier 2 provides primary inference.
3. **LGCM-lmer convergent evidence** failed the pre-registered $|\Delta| < 0.005$ threshold for all grades. Sensitivity analysis attributes this to method (FIML linear-growth imputation vs REML on observed cases) rather than nonlinear growth — interpreted as a substantive method effect, not a measurement problem.
4. **CT-DSEM uses MAP estimation** (optim mode) for point estimates with Wald-type CIs. Full Bayesian MCMC remains as future sensitivity check.
5. **2PL EAP scoring not done** as primary; 1PL Rasch was used due to 2-6% degenerate items ($a \leq 0$ or $|b| > 5$) in 2PL fits. 2PL kept for future sensitivity.
6. **Phase 1 only:** This report covers school year 2024-2025. Cross-year linking and trends are out of scope.
7. **Per-grade calibration:** No cross-grade theta scale linking. Inferences within grades only — cross-grade comparisons are qualitative (pattern shifts), not quantitative magnitude comparisons.
8. **No teacher/class metadata:** the OLM math collections (`math_teacher_static`, `math_teacher_categories`, `math_teacher_course`) do not contain a `class_id` / `teacher_id` / `school_id` field, so the proposal’s MT4(c) *teaching-effectiveness* analysis is deferred to Phase 2 pending schema enrichment. See `docs/scope_refinement.md`.
9. **Single-construct measurement → no RI-CLPM:** the proposal lists RI-CLPM among four LSEM variants, but RI-CLPM requires 2 constructs measured longitudinally to identify cross-lagged effects; Phase 1 has only mathematics ability, so RI-CLPM is over-specified and was dropped on methodological grounds rather than for lack of time.
10. **VI/OIRT delivered as a design contribution only:** the streaming variational form of MT2 needs a deployment platform (the proposal’s §8.3 production code is also deferred). The algorithmic specification — ELBO, streaming update, complexity — is documented in `docs/oirt_vi_design.md`.

13. Conclusion

Integrating IRT with longitudinal SEM enables tracking of *dynamic* math ability on a Vietnamese online learning system. Across grades 10–12 we find consistent, small, positive growth (RQ1) and a robust **catch-up → plateau shift** (RQ3): a compensatory regime in Grade 10 that weakens through Grade 11 and disappears by Grade 12, corroborated by a continuous-time mean-reversion model. The dynamics are directionally consistent across grades and estimation methods (RQ2), while the differences in magnitude are substantively interpretable rather than artefacts. These results carry a direct pedagogical message — intervene early, while compensatory dynamics still help — and lay the methodological groundwork for Phase 2 (cross-year vertical scaling with the 2025-2026 review course as natural anchor items).

Two downstream applications translate the dynamic estimates into actionable signals for the OLM platform: a real-time Kalman smoothing layer (B6) that tightens per-occasion ability estimates by absorbing measurement noise across time, and prototype early-warning + item-recommendation tools (B7) that surface at-risk students and serve information-maximising next items.

14. Reproducibility

Pipeline scripts in `research/irt_lsem/src/r/`:

	Script	Step	Runtime
b1_dimensionality.R	B1		~30 min
b2_calibration.R	B2		~3 hours
b2_postprocess.R	B2 cleanup		~5 min
b3_eap_scoring.R	B3		~20 min
b4a_lgcm_lscsm.R	B4a		~17 min
b4a_test.R	B4a sensitivity		~2 min
b4c_ctdsem.R	B4c		~2 min (optim)
b4c_viz.R	B4c plots		~1 min
b5_consolidate.R	B5 tables		<1 min
b5_plots.R	B5 figures		<1 min
b6_kalman.R	B6 KFAS smoother		~2 min
b6_viz.R	B6 plots		<1 min
b7a_early_warning.R	B7a flagging		<1 min
b7b_recommendation.R	B7b recommender		<1 min
b5_report.Rmd	this document		<1 min

Outputs are CSV/RDS/PNG; all artefacts in `outputs/`. R version R version 4.5.2 (2025-10-31). Random seed = 42 across pipeline.

```
## R version 4.5.2 (2025-10-31)
## Platform: x86_64-pc-linux-gnu
## Running under: Debian GNU/Linux 12 (bookworm)
##
## Matrix products: default
## BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p0.3.21.so; LAPACK version 3.11.0
##
## locale:
## [1] LC_CTYPE=en_US.UTF-8 LC_NUMERIC=C
## [3] LC_TIME=en_US.UTF-8 LC_COLLATE=en_US.UTF-8
## [5] LC_MONETARY=en_US.UTF-8 LC_MESSAGES=en_US.UTF-8
## [7] LC_PAPER=en_US.UTF-8 LC_NAME=C
## [9] LC_ADDRESS=C LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Asia/Bangkok
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats graphics grDevices utils datasets methods base
##
## other attached packages:
## [1] knitr_1.50 data.table_1.17.6
##
## loaded via a namespace (and not attached):
## [1] bit_4.6.0 compiler_4.5.2 fastmap_1.2.0 cli_3.6.5
## [5] tools_4.5.2 htmltools_0.5.8.1 yaml_2.3.10 bit64_4.6.0-1
## [9] rmarkdown_2.29 xfun_0.52 digest_0.6.37 rlang_1.1.6
## [13] evaluate_1.0.4
```